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CHM 191 — Past Questions & Materials

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Course: CHM 191

Institution: Federal University of Technology, Minna

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Paper 1 of 1 — CHM191 report

Level 100 • First Semester • PastQuestion

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Experiment 1

TITLE: DETERMINATION OF THE AMOUNT OF NaOH AND Na₂CO₃ IN A MIXTURE.

AIM: TO DETERMINE THE AMOUNT OF NaOH AND Na₂CO₃ IN A MIXTURE.

Theory/Introduction: Mixed Titration involving two alkali in a mixture against an acid. Titration of HCl against a mixture, solution 'B' containing NaOH and Na₂CO₃ to determine the amount of each of the substances in solution 'B' using two indicators.

Reactions:

$$\text{HCl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$$

$$\text{HCl} + \text{Na}_2\text{CO}_3 \longrightarrow \text{NaHCO}_3 + \text{NaCl}$$

$$\text{HCl} + \text{NaHCO}_3 \longrightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$$

Materials:

50cm³ Burette, 25cm³ Pipette, 250cm³ Conical flask, Retort stand, with clamp, wash bottle, Beaker (250cm³), Methyl orange and Phenolphthalein indicator, 'A' is 0.05M of HCl and 'B' is a solution of NaOH and Na₂CO₃.

Procedure:

i. Rinsed the burette with standard solution of HCl

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before filling the burette with the standard solution of hydrochloric acid (HCl) and then adjusted the burette to its zero mark.

ii. Pipetted 25cm^3 of solution 'B' into a conical flask and two drops of phenolphthalein indicator was added, and then the solution was titrated against the acid, swirling the flask at each time to evenly mix the two solutions and the pink colour of solution 'B' finally disappears, and the solution became colourless, the volume of acid used was noted.

iii. Without discarding the solution, two drops of methyl orange was added to the solution, and then the solution was again titrated against the acid, swirling the flask continuously until the end point is reached, which is when the solution turns orange, the second volume of acid was also noted,

iv. The processes were carried out two more times to have an accurate reading, and the burette readings were recorded.

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Table^o

Titration (cm ³)	Rough	1st titre	2nd titre	3rd titre
2nd end Point (V ₂)	25.30	25.20	25.40	39.10
1st end Point (V ₁)	15.40	15.20	15.40	29.00
Initial volume (V ₀)	0.00	0.00	0.00	13.70
Volume (V ₁ - V ₀)	15.40	15.20	15.40	15.30
Volume (V ₂ - V ₁)	9.90	10.00	10.00	10.10
Volume (V ₂ - V ₀)	25.30	25.20	25.40	25.40

Calculations^o

$$\text{Average volume of acid used (POP)} = \frac{15.20 + 15.40 + 15.30}{3}$$

$$= 15.30 \text{ cm}^3$$

$$\text{Average volume of acid used (MO)} = \frac{10.00 + 10.00 + 10.10}{3}$$

$$= 10.03 \text{ cm}^3$$

$$\text{Total average of volume of acid used} = \frac{25.20 + 25.40 + 25.40}{3}$$

$$= 25.33 \text{ cm}^3$$

i) The volume of HCl required for complete neutralization of Na₂CO₃ = average total volume of acid used - Volume of acid reacting with NaOH

$$= [V_2 - (x - y)] \text{ cm}^3$$

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Where x = burette reading at Phenolphthalein end Point.

Where y = burette reading at methyl orange end Point minus burette reading at Phenolphthalein end Point.

$$= [25.33 - (15.30 - 10.03)]$$

$$= [25.33 - 5.27]$$

$$= 20.06 \text{ cm}^3$$

ii) Volume of acid required for complete neutralization of NaOH = $(x - y) \text{ cm}^3$

$$= [15.30 - (25.33 - 15.30)]$$

$$= [15.30 - 10.03]$$

$$= 5.27 \text{ cm}^3$$

iii) Conc. (mol/dm^3) of solution NaOH that has reacted

$$\frac{C_A V_A}{C_B V_B} = \frac{n_A}{n_B} \quad C_A = 0.05 \text{ mol/dm}^3, V_A = 5.27 \text{ cm}^3$$

$$C_B V_B \quad n_B \quad C_B = ?, V_B = 25.00, n_A = 1, n_B = 1$$



$$0.05 \times 5.27 = \frac{1}{1}$$

$$C_B \times 25.00 = 1$$

$$C_B = \frac{0.05 \times 5.27}{25.00}$$

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$$C_B = 0.011 \text{ mol/dm}^3$$

iv) Conc. (mol/dm³) of solution Na₂CO₃ in the mixture

$$C_A = 0.05 \text{ mol/dm}^3, V_A = 20.06 \text{ cm}^3, n_A = 1, C_B = ?, V_B = 25.00 \text{ cm}^3$$

$$n_B = 1$$

$$\frac{C_A V_A}{C_B V_B} = \frac{n_A}{n_B} ; \frac{0.05 \times 20.06}{C_B \times 25.00} = \frac{1}{1}$$

$$C_B = \frac{1.003}{25.00} = 0.04 \text{ mol/dm}^3$$

v.) Amount of NaOH in grams per dm³:

$$\begin{aligned} \text{Conc. (g/dm}^3) &= \text{molar mass} \times \text{Conc. (mol/dm}^3) \\ &= (23 + 16 + 1) \times 0.011 \\ &= 40 \times 0.011 \\ &= 0.44 \text{ g/dm}^3 \end{aligned}$$

vi) The amount in grams per dm³ of the Na₂CO₃ in the mixture.

$$\begin{aligned} \text{Conc. (g/dm}^3) &= \text{molar mass} \times \text{Conc. (mol/dm}^3) \\ &= (23 \times 2 + 12 + 16 \times 3) \times 0.04 \\ &= 106 \times 0.04 \\ &= 4.24 \text{ g/dm}^3 \end{aligned}$$

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vii) The Percentage Composition of NaOH in the mixture

$$\% \text{ Comp of NaOH} = \frac{\text{No of moles of NaOH (g/mol)}}{\text{Total number of moles (g/mol)}} \times 100$$

$$= \frac{0.44}{0.44 + 2.24} \times 100$$

$$= \frac{0.44}{4.68} \times 100 = 9.40\%$$

viii) The Percentage composition of Na_2CO_3 in the mixture

$$\% \text{ Comp. } \text{Na}_2\text{CO}_3 = \frac{4.24}{4.68} \times 100$$

$$= 90.598 \approx 90.6\%$$

Questions:

- 1) A mixed titration is a type of titration in which two acids or two base reagents in a solution are titrated against a single base or acid in order to determine their respective quantities in the mixture.
- 2) It is not recommended to use a single indicator for mixed titration because it involves either two acid or base mixture, and since the two acids or bases in the mixture have different pH, different

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Indicators are required to acquire their endPoints simultaneously.

3.

Single Reagent	Mixed Reagent
This involves a single base ^{or acid} against another base or acid.	This involves two bases or acids against one base or acid.
* This requires one indicator	This requires two or more indicators.

Discussion:

While using the Phenolphthalein indicator, the pink colour was observed and after titrating against the acid the colour disappears, leaving the solution colourless, indicating the first reaction between NaOH and HCl , and after adding the methyl orange, another colour change was noticed and after titrating against the acid one more time, the end point was reached and the solution became orange, indicating that Na_2CO_3 and NaHCO_3 in the mixture has been neutralized.

Precautions:

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- i) I ensured proper rinsing of my burette with a standard solution of the acid.
- ii) I avoided air bubbles while filling my burette and pipette.
- iii) I ensured to remove the filter funnel from the burette once it is filled to zero mark.
- iv) I avoided error due to parallax while reading my burette.
- v) I ensured to place a white paper underneath my conical flask while titrating.

Conclusion:

After the titrations, it was clear that there was more of NaOH in the mixture than there is of Na_2CO_3 . There is about 90.6% of NaOH contained in the mixture compared to Na_2CO_3 of 9.40%.

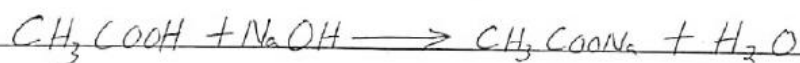
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Experiment 2

Title: Determination of the concentration of a weak acid in the presence of a strong acid in the mixture using a strong base.

Aim: To determine the concentration of HCl and CH_3COOH in a mixture.

Theory/Introduction: It is a reaction between a mixture containing a strong acid 'HCl' and weak acid ' CH_3COOH ' against a strong base ' NaOH ' to give the following equations.



Materials:

50cm³ burette, 25cm³ pipette, Retort stand with clamp, conical flask, 250cm³ Beaker (250cm³), 'A' is a mixture of a weak acid (CH_3COOH) and strong acid (HCl), 'B' is a solution NaOH, methyl orange and Phenolphthalein indicator.

Procedure:

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i) The burette was properly rinsed with a standard solution of NaOH before filling the burette with NaOH till it reaches zero mark on the burette, and noted the initial reading.

ii) 25cm^3 of the acid mixture was pipetted into the conical flask, then two drops of methyl orange was added to the acid mixture and it was titrated against the NaOH solution, a colour change was observed from ~~pink~~^{pink} to yellow, indicating the first end point, and the burette reading was noted.

iii) Without discarding the solution, or adding base to the burette, or more acid into the conical flask, Two drops of Phenolphthalein was added to the same solution, and then titrated against the NaOH solution, swirling the conical flask until a colour change was observed from yellow to orange. Note the second burette reading at Phenolphthalein end point, and repeat the steps again in order to get an average titre value.

Table:

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Burette readings (cm ³)	Rough	1st titre	2nd titre	3rd titre
Second end Point (V ₂)	27.30	26.90	27.10	27.10
First end Point (V ₁)	21.50	21.30	21.30	21.40
Initial Volume (V ₀)	0.00	0.00	0.00	0.00
Volume of base used (V ₂ -V ₁)	21.50	21.30	21.30	21.40
Volume of base used (V ₂ -V ₀)	5.80	5.60	5.80	5.70
Volume of base used (V ₁ -V ₀)	27.30	26.90	27.10	27.10
Average Volume of base used for M ₁ =				
$= \frac{21.30 + 21.30 + 21.40}{3} = 21.33 \text{ cm}^3$				
Average Volume of base used for P ₁ =				
$= \frac{5.60 + 5.80 + 5.70}{3} = 5.70 \text{ cm}^3$				
Average Volume of acid used for the total acid mix				
$\text{line} = \frac{26.90 + 27.10 + 27.10}{3} = 27.03 \text{ cm}^3$				
Calculations:				
Volume of NaOH required for complete neutralization of HCl = Volume of NaOH used at methyl-orange end-point				

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i) Volume of NaOH required for complete neutralization of HCl = 21.33 cm^3

ii) Volume of NaOH required for complete neutralization of the total acid mixture = 27.03 cm^3

iii) Volume of NaOH required for complete neutralization of CH_3COOH in the mixture - Volume of NaOH used at Phenolphthalein end Point = 5.70 cm^3

iv) Conc. (mol/dm^3) of HCl in the mixture

$$\frac{C_A V_A}{C_B V_B} = \frac{n_A}{n_B} \quad C_A = ?, V_A = 25.00 \text{ cm}^3, C_B = 0.15 \text{ M}, V_B = 21.33 \text{ cm}^3$$

$$n_A = 1, n_B = 1$$

$$\frac{C_A \times 25.00}{0.15 \times 21.33} = \frac{1}{1} \quad ; \quad C_A = \frac{0.15 \times 21.33}{25.00}$$

$$C_A = 0.1279 \approx 0.13 \text{ mol/dm}^3$$

v) Conc. (mol/dm^3) of CH_3COOH

$$C_A = ?, C_B = 0.15 \text{ M}, V_A = 25.00 \text{ cm}^3, V_B = 5.70 \text{ cm}^3, n_A = 1$$

$$n_B = 1 \quad \frac{C_A V_A}{C_B V_B} = \frac{n_A}{n_B}$$

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$$C_A \times 25.00 = 1$$

$$0.15 \times 5.70 = 1$$

$$C_B = \frac{0.15 \times 5.70}{25.00} = 0.0342 \approx 0.034 \text{ mol/dm}^3$$

Vi) Concentration (g/dm³) of HCl in the mixture

$$\text{Conc. (g/dm}^3\text{)} = \text{Conc. (mol/dm}^3\text{)} \times \text{molar mass}$$

$$\text{Conc. (g/dm}^3\text{)} \text{ of HCl} = 0.13 \times (1 \times 1 + 35.5 \times 1)$$

$$= 0.13 \times 36.5$$

$$= 4.745 \approx 4.75 \text{ g/dm}^3$$

Vii) Concentration of CH₃COOH in g/dm³

$$\text{Conc. (g/dm}^3\text{)} = \text{Conc. (mol/dm}^3\text{)} \times \text{molar mass (g/mol)}$$

$$= 0.034 \times (12 \times 2 + 4 \times 1 + 16 \times 2)$$

$$= 0.034 \times (24 + 4 + 32)$$

$$= 0.034 \times 60$$

$$= 2.04 \text{ g/dm}^3$$

Viii) Percentage Composition of HCl in the mixture

$$\% \text{ HCl} = \frac{4.75}{2.04 + 4.75} \times 100 = \frac{4.75}{6.79} \times 100$$

$$2.04 + 4.75$$

$$6.79$$

$$= 69.95\% \text{ of HCl in the mixture}$$

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ix) Percentage Composition of CH_3COOH in the mixture

$$\% \text{CH}_3\text{COOH} = \frac{2.04}{4.75 + 2.04} \times 100$$

$$= \frac{2.04}{6.79} \times 100$$

$$= 30.04\% \text{ of } \text{CH}_3\text{COOH} \text{ in the mixture}$$

x) Calculate the pH of the acid mixture



$$\text{H}^+ \text{ ion concentration } [\text{H}^+] = [1 \times 0.13] = 0.13 \text{ mol/dm}^3$$



$$\text{H}^+ \text{ ion concentration } [\text{H}^+] = [1 \times 0.034] = 0.034 \text{ mol/dm}^3$$

Total concentration of H^+ in the mixture

$$= (0.13 + 0.034) \text{ mol/dm}^3 = 0.164 \text{ mol/dm}^3$$

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\text{pH} = -\log_{10} [0.164]$$

$$\text{pH} = -(-0.7851)$$

$$\text{pH} = 0.785$$

$$\text{pH} \approx 0.79$$

Discussion:

In the titration involving weak acids and a base

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In order to determine the concentration of the weak acid (CH_3COOH), the use of two indicators was abridged and Phenolphthalein was the indicator used for the neutralization of the CH_3COOH in the mixture eventually using the titre values obtained to find the amount of CH_3COOH and HCl in the mixture.

Precautions:

- I ensured to rinse the burette with a small amount of Standard Solution of NaOH .
- I avoided air bubbles while filling the burette and Pipette.
- I ensured to remove the filter funnel from the burette once it is filled to its zero mark.
- I ensured to read the burette readings from the lower meniscus.
- I avoided Parallax error while reading the burette.

Conclusion:

The use of two indicators was required, in order to determine the amount of HCl and CH_3COOH in the mixture, in other words, without the use of the suitable

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Indicators, It would have been impossible to determine the respective concentrations of HCl and CH_3COOH in the mixture. After the experiment and proper calculations, it was observed that there is more of HCl (about 89.95%) in the mixture than CH_3COOH (about 30.04%) in the mixture.

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Experiment 3

Title:

Determination of the Partition/Distribution Coefficient of a Solute (Iodine) between two immiscible liquids.

Aim:

To determine the Partition/Distribution Coefficient of Iodine between Chloroform and water.

Theory/Introduction:

It is a reaction of two immiscible solvents (water and chloroform) in the presence of a solute (Iodine).

Iodine which is soluble in both water and chloroform is added to the solvents. Iodine will be distributed among the two solvents until a dynamic equilibrium is achieved. At this state, the state of Iodine molecules in both solvents 'Chloroform and water' is the same.

Materials:

Burette (50 cm^3), Conical flask (250 cm^3), 25 cm^3 Pipette, Solid Iodine, Separatory funnel, distilled water, Starch Indicator, 10% Potassium Iodide (KI) solution, Sodium thiosulphate solution.

Procedure:

- i) 0.7g of solid Iodine was weighed into a beaker
- ii) 12 cm^3 of Chloroform was put into a dry stoppered Separatory funnel, and then weighed 0.7g of solid Iodine

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was added into the chloroform in the separator funnel, and then 10cm^3 of distilled water was added to the iodine and chloroform and the separator funnel was covered and shaken for about 20 to 30 minutes until a dynamic equilibrium was established (i.e., iodine was distributed among the chloroform and water), then the mixture was allowed to stand for 5-10 minutes until the phase was completely separated into its different layers.

(iii) Separated the organic layer from the aqueous layer.

(iv) 2.5cm^3 of the organic layer was measured and transferred into a conical flask, and 10cm^3 of 10% KI was added, and then titrated against a standard Sodium thiosulphate solution while shaking the flask until it became pale yellow, then two drops of starch indicator was added and titration was continued until the blue-black colour of the solution disappeared.

(v) 25cm^3 of the aqueous layer was pipetted into another conical flask, and two drops of starch indicator was added and then titrated against the standard Sodium thiosulphate solution until the blue-black

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Colour disappears.

Table:

Organic layer

Burette reading (cm^3)	Reading
Organic layer + 10cm^3 of 10% KI + Starch (V)	12.30
Organic layer + 10cm^3 of 10% KI (V)	11.80
Initial reading (V_0)	0.00
Total volume of $\text{Na}_2\text{S}_2\text{O}_3$ used ($V_1 - V_0$)	12.30

Aqueous layer titrated against $\text{Na}_2\text{S}_2\text{O}_3$

Burette readings (cm^3)	Reading
Final volume of $\text{Na}_2\text{S}_2\text{O}_3$ used (V_1)	13.60
Initial volume (V_0)	12.30
Total volume used ($V_1 - V_0$)	1.30

Calculations:

Calculate the concentration of iodine in the aqueous layer.

Conc of $\text{Na}_2\text{S}_2\text{O}_3 \times \text{Vol. of } \text{Na}_2\text{S}_2\text{O}_3 = \text{Conc. of iodine in aqueous layer} \times \text{Volume of aqueous layer.}$

$$0.1 \text{ mol/dm}^3 \times 1.30 \text{ cm}^3 = \text{Conc. of } \text{I}_2 \times 25 \text{ cm}^3$$

$$C_I = \frac{0.1 \times 1.30}{25} = \frac{0.13}{25.00}$$

$$= 0.0052 \text{ mol/dm}^3$$

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ii) Calculate the concentration of Iodine in the organic layer.

Conc. of $\text{Na}_2\text{S}_2\text{O}_3 \times \text{Vol. of } \text{Na}_2\text{S}_2\text{O}_3 = \text{Conc. of } \text{I}_2 \text{ in organic layer} \times \text{Volume of } \text{I}_2 \text{ in organic layer}$

$$0.1 \text{ mol/dm}^3 \times 12.30 \text{ cm}^3 = C_{\text{I}_2} \times 2.5 \text{ cm}^3$$

$$C_{\text{I}_2} = \frac{0.1 \times 12.30}{2.5} = 1.23$$

$$C_{\text{I}_2} = 0.492 \text{ mol/dm}^3$$

iii) Calculate the Partition Coefficient

$$K_D = \frac{0.492}{0.0052} = 94.61$$

Questions:

1) Partition Coefficient (ρ) is the ratio of concentrations of a compound in a mixture of two immiscible phases at equilibrium.

2) Partition Coefficient, $K_D = \frac{\text{Concentration of X in Liquid A}}{\text{Concentration of X in Liquid B}}$

3) A solid X is added to a mixture of benzene and water, after shaking well and allowed to stand

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10cm^3 of the benzene layer was found to contain 0.13g of X and 100cm^3 of water layer contained 0.22g of X . Calculate the value of the distribution Coefficient.

Solution

If $10\text{cm}^3 \longrightarrow 0.13\text{g}$

$1000\text{cm}^3 \longrightarrow x\text{g}$

$$x = \frac{1000\text{cm}^3 \times 0.13\text{g}}{10\text{cm}^3} = 13\text{g/dm}^3$$

also, if 100cm^3 of $\text{H}_2\text{O} \longrightarrow 0.22\text{g}$

$1000\text{cm}^3 \longrightarrow y\text{g}$

$$y = \frac{1000 \times 0.22}{100} = 2.2\text{g/dm}^3$$

Concentration of X in benzene $= 13\text{g/dm}^3$

While concentration of X in water $= 2.2\text{g/dm}^3$

$$K_D = \frac{13}{2.2} = 5.9$$

6) At 25°C , an aqueous solution of iodine containing 0.0516g/dm^3 is in equilibrium with a carbon tetrachloride (CCl_4) solution containing 4.412g/dm^3 . The solubility of iodine in water at 25°C is 0.34g/dm^3 . find the solubility of

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of Iodine in Carbon tetrachloride.

Solution

$K_D = \frac{\text{Conc. of Iodine in Carbon tetrachloride (CCl}_4\text{)}}{\text{Conc. of Iodine in aqueous solution}}$

$$\frac{4.412}{0.0516} = \frac{x}{0.34}$$

$$0.0516 \quad 0.34$$

$$x = \frac{4.412 \times 0.34}{0.0516} = 1.50008$$

$$0.0516 \quad 0.0516$$

$$x = 29.079 \text{ g/dm}^3$$

In other words, 29.079 g of Iodine will dissolve in 1 dm³ of Carbon tetrachloride (CCl₄).

7) The Solubility of an organic Solute in hexane is 7.5 g/100 cm³, the Solubility of the Solute in water is 1.5 g/50 cm³, Calculate K_D .

Solution

$K_D = \frac{\text{Conc. of the Solute in hexane}}{\text{Conc. of the Solute in water}}$

Conc. of the Solute in water

Conc. of Solute in hexane = 7.5 g in 100 cm³

While Conc of Solute in water = 1.5 g in 50 cm³

is 1.5 g $\rightarrow$ 50 cm³

7.5 g $\rightarrow$ 100 cm³

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$$x = \frac{1.5 \times 10^2}{5.0}$$

5.0

$$x = 3.9 \text{ in } 100 \text{ cm}^3$$

$$k_D = \frac{7.5}{3} = 2.5$$

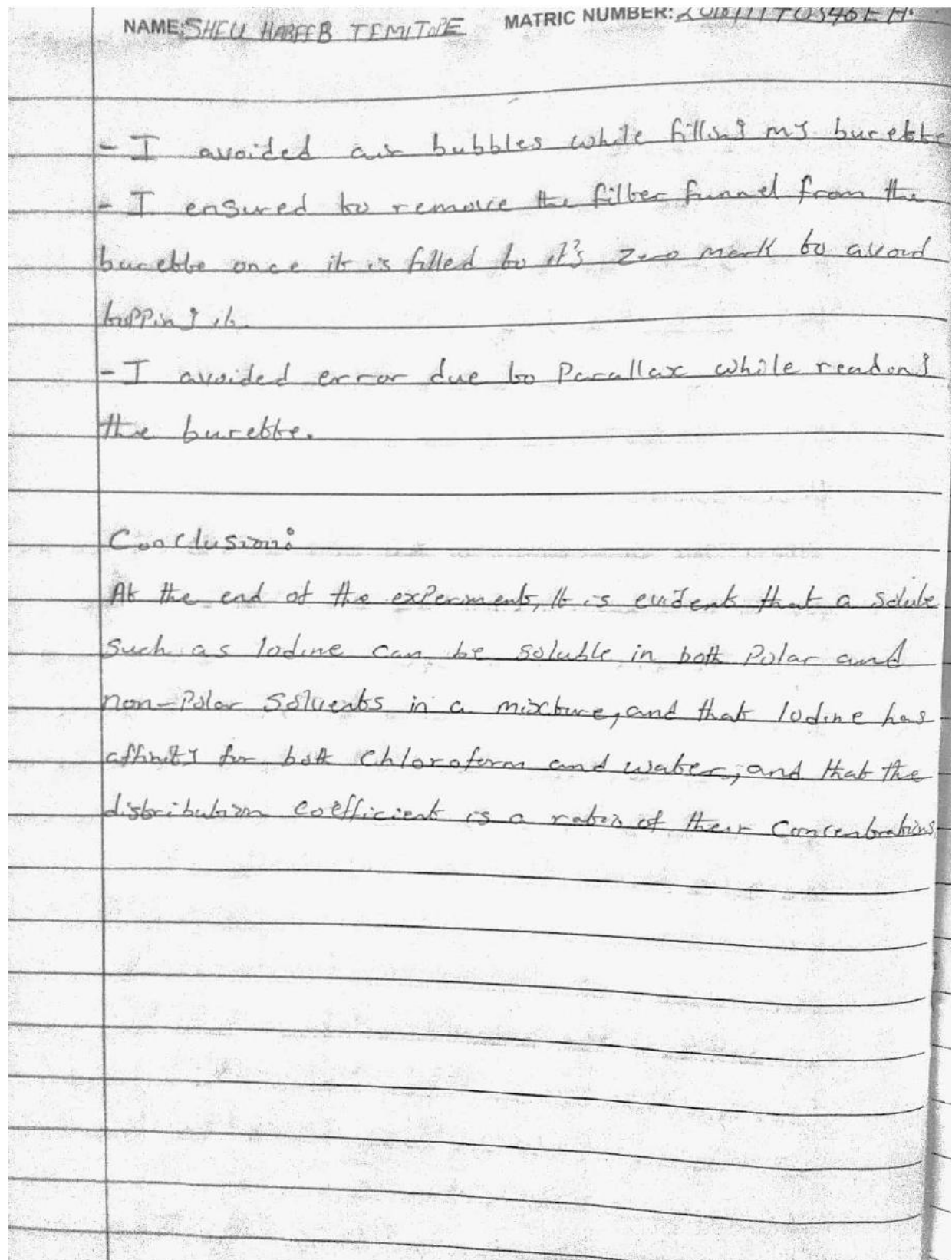
Discussion:

After the separation of the organic layer and the aqueous layer, 10 cm^3 of KI was added to the organic because $\text{Na}_2\text{S}_2\text{O}_3$ is Polar Solvent, and therefore won't react with Trichloromethane, so 10 cm^3 of 10% Potassium Iodide was added to dissolve the iodine in the organic layer, before titrating it against $\text{Na}_2\text{S}_2\text{O}_3$.

Precautions:

- I ensured that a dynamic equilibrium was achieved between the organic layer and aqueous layer.
- I ensured to let the mixture in the separator funnel to stand before separating it.
- I ensured to rinse the burette with the standard solution of $\text{Na}_2\text{S}_2\text{O}_3$ before filling it with $\text{Na}_2\text{S}_2\text{O}_3$.

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NAME: Shau Habeeb Tensitabe MATRIC NUMBER: 201811170346EHExperiment 4Title:Solubility Properties of Compounds.Aim:To determine the Solubility Properties of some Compounds.Theory/Introduction:

The Solubility of a substance is defined as the mass of that substance which will dissolve in a certain amount of solvent. Compounds or substances that are soluble in a particular solvent is said to have affinity for that solvent. Some compounds are soluble in a particular or specific solvents while others are not.

Materials:

Conical flask (250cm^3), measuring cylinder (100cm^3), wash bottle, Beaker (250cm^3), Distilled water, Test tube, different inorganic salts (A, B, C, D, and E).

Procedure:

- * All glasswares such as test tube, measuring cylinder and conical flask were all rinsed with distilled water.
- * All inorganic salts 'A, B, C, D and E' were put in separate test tubes labelled A, B, C, D and E respectively.

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* 2cm^3 of distilled water was added to test tube A, C, and D, then 3cm^3 of distilled water was added to B while 10cm^3 was added to F, all test tubes were swirled and allowed to stand to observe the solubility and colour of each of the solution.

* All observations were recorded into a table.

Table 1:

Test	Colour of Solids	Solubility in water	Colour of solution
A	White	Soluble	Colourless
B	White	Insoluble	Colourless
C	White	Soluble	Colourless
D	Crystalline white	Soluble	Colourless
E	White Powder	Sparingly Soluble	White precipitate

Questions 1:

Salt	Soluble	Insoluble
AgSO_4	Soluble	
CaCO_3		Insoluble
Na_3PO_4	Soluble	
$\text{Zn}(\text{NO}_3)_2$	Soluble	

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Question 2:

Solubility is the amount of substance or solute that can dissolve in a particular or specific volume of solvents at a given temperature.

i) Slightly Soluble means the substance can't fully dissolve in certain volume of the solvent at a given temperature.

ii) Weak electrolyte is a solution of salt that can't fully dissociate into its ionic form in the solution.

iii) A solution is said to be saturated when it contains as much solute as it can dissolve at a particular temperature.

iv) A super saturated solution is one which contains more solute than it can dissolve at that particular temperature.

Discussion:

After dissolving each of the salts in water, it could be said that salt A, C, D are ionic salts of chlorides, common hexamoxosulphate (vi) salts or

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TrioxoCarbonate of Na, K and NH_4 , while B could be trioxoCarbonate(IV) of element such as Calcium or trioxoSulphate(IV) of other elements excluding Na, K, NH_4 , and F. Could be said to be tetroxoSulphate(VI) of Calcium or Lead.

Precautions:

- * I avoided Parallax error while reading the measuring cylinder.
- * I ensured that ~~the~~ measuring cylinder was properly rinsed with distilled water.
- * I ensured that the salts were all covered before I start my experiment in order to avoid deliquescence.

Conclusion:

After the whole experiment, it is obvious that not all Insoluble Salts are actually soluble in water. Hence, all the Insoluble Salts of Na, K, NH_4 are soluble in water.

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